

Discrete Time Fourier Transform (DTFT)

→ For Discrete time aperiodic signal

→ Aperiodic signal $\equiv$ Periodic signal ($N \rightarrow \infty$)

$$\begin{aligned} x[n] &\longleftrightarrow X(\omega) \\ X(\omega) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \\ x[n] &= \frac{1}{2\pi} \int_{2\pi} X(\omega) e^{j\omega n} d\omega \end{aligned}$$

→ DTFT, $X(\omega) \Rightarrow$ Continuous function of ω

$\Rightarrow$ Periodic in ω with period 2π

$\Rightarrow$ Spectrum of $x[n]$

Q. $x[n] = a^n u[n]$; $|a| < 1$. Find DTFT

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} a^n e^{-j\omega n}$$

$$= 1 + a e^{-j\omega} + a^2 e^{-j2\omega} + \dots + a^{\infty} e^{-j\omega \infty}$$

$$X(\omega) = \frac{1}{1 - a e^{-j\omega}} \quad \left(\begin{array}{l} \text{Sum of infinite G.P.} \\ |a e^{-j\omega}| < 1 \end{array} \right)$$

$$|X(\omega)| = \frac{1}{1}$$

$$X(\omega) = \frac{1}{1 - a(\cos\omega - j\sin\omega)}$$

$$|X(\omega)| = \frac{1}{\sqrt{(1 - a\cos\omega)^2 + (a\sin\omega)^2}}$$

$$= \frac{1}{1 - a\cos\omega + j a\sin\omega}$$

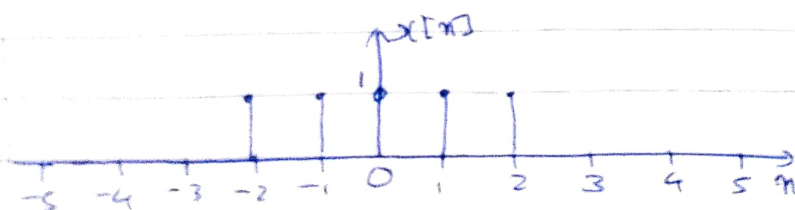
$$\angle X(\omega) = -\tan^{-1} \left(\frac{a\sin\omega}{1 - a\cos\omega} \right)$$

Q $x[n] = a^n$; $|a| < 1$ Find DTFT

$$\begin{aligned}
 X(\omega) &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \\
 &= \sum_{n=-\infty}^{-1} a^{-n} e^{-j\omega n} + \sum_{n=0}^{\infty} a^n e^{-j\omega n} \\
 &= \left[a^1 e^{+j\omega} + a^2 e^{+j\omega 2} + a^3 e^{+j\omega 3} + \dots \right] + \left[1 + a e^{-j\omega} + a^2 e^{-j\omega 2} + \dots \right] \\
 &= \frac{a e^{j\omega}}{1 - a e^{j\omega}} + \frac{1}{1 - a e^{-j\omega}} \\
 &= \frac{1}{a^{-1} e^{-j\omega} - 1} + \frac{1}{1 - a e^{-j\omega}} \\
 &= \frac{1}{1 - a e^{-j\omega}} - \frac{1}{1 - a^{-1} e^{-j\omega}} \\
 &= \frac{(1 - a^{-1} e^{-j\omega}) - (1 - a e^{-j\omega})}{(1 - a e^{-j\omega})(1 - a^{-1} e^{-j\omega})} = \frac{1 - a^{-1} e^{-j\omega} - 1 + a e^{-j\omega}}{1 - a^{-1} e^{-j\omega} - a e^{-j\omega} + e^{-j\omega 2}} \\
 &= \frac{-a^{-1} + a}{e^{j\omega} - a^{-1} - a + e^{-j\omega}} = \frac{a - \frac{1}{a}}{2\left(\frac{e^{j\omega} + e^{-j\omega}}{2}\right) - \frac{1}{a} - a} = \frac{a - \frac{1}{a}}{2 \cos \omega - \frac{1}{a} - a}
 \end{aligned}$$

$$X(\omega) = \frac{a^2 - 1}{2a \cos \omega - 1 - a^2} \rightarrow \text{Purely a real quantity}$$

Q $x[n] = \begin{cases} 1 & ; |n| \leq 2 \\ 0 & ; \text{otherwise} \end{cases}$; Find DTFT



$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$= \sum_{n=-2}^2 (1) e^{-j\omega n} = e^{j\omega 2} + e^{j\omega} + 1 + e^{-j\omega} + e^{-j\omega 2}$$

$$= 2 \left[\frac{e^{j\omega 2} + e^{-j\omega 2}}{2} \right] + 2 \left[\frac{e^{j\omega} + e^{-j\omega}}{2} \right] + 1$$

$$= 2 \cos 2\omega + 2 \cos \omega + 1$$

$$= \frac{(2 \cos 2\omega + 2 \cos \omega + 1) \sin(0.5\omega)}{\sin(0.5\omega)}$$

$$= \frac{2 \cos 2\omega \sin(0.5\omega) + 2 \cos \omega \sin(0.5\omega) + \sin(0.5\omega)}{\sin(0.5\omega)}$$

$$= \frac{\sin(2.5\omega) - \sin(1.5\omega) + \sin(1.5\omega) - \sin(0.5\omega) + \sin(0.5\omega)}{\sin(0.5\omega)}$$

$$X(\omega) = \frac{\sin(2.5\omega)}{\sin(0.5\omega)}$$

⇒ Convergence of DTFT :- $x[n]$ is absolutely summable

$$\sum_{n=-\infty}^{\infty} |x[n]| < \infty$$

⇒ Properties of DTFT

$$x[n] \longleftrightarrow X(\omega)$$

$$\bullet \quad X(\omega + 2\pi) = X(\omega) \quad \left(\begin{array}{l} \text{Periodicity of DTFT} \\ \text{Period} = 2\pi \end{array} \right)$$

$$\bullet \quad a x_1[n] + b x_2[n] \longleftrightarrow a X_1(\omega) + b X_2(\omega) \quad (\text{Linearity})$$

$$\bullet \quad x[n - n_0] \longleftrightarrow e^{-j\omega n_0} X(\omega) \quad (\text{Time shifting})$$

- $e^{j\omega_0 n} x[n] \longleftrightarrow X(\omega - \omega_0)$
(Frequency Shifting)

- $x^*[n] \longleftrightarrow X^*(-\omega)$
(Conjugate Symmetry)

- $x[-n] \longleftrightarrow X(-\omega)$ (Time Reversal)

- $x\left[\frac{n}{K}\right] \longleftrightarrow X(K\omega)$
(Time expansion)

$$x\left[\frac{n}{K}\right] = \begin{cases} x\left[\frac{n}{K}\right] & ; \text{ if } n \text{ is multiple of } K \\ 0 & ; \text{ if } n \text{ is not multiple of } K \end{cases}$$

- $n x[n] \longleftrightarrow j \frac{dX(\omega)}{d\omega}$ (Differentiation in Frequency)

- $x[n] * h[n] \longleftrightarrow X(\omega) H(\omega)$
(Convolution property)

- $x_1[n] x_2[n] \longleftrightarrow \frac{1}{2\pi} \int_{2\pi} X_1(\tau) X_2(\omega - \tau) d\tau$
(Multiplication property)

- $\sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{2\pi} |X(\omega)|^2 d\omega$
(Parseval's Relation)

- There is no duality (or symmetric) property in case of DTFT

Q

$$x[n] \leftrightarrow X(\omega)$$

$$x[n-n_0] \leftrightarrow ?$$

$$F[x[n-n_0]] = \sum_{n=-\infty}^{\infty} x[n-n_0] e^{-j\omega n}$$

$$\text{let } n-n_0 = n_1 \Rightarrow n = n_1 + n_0$$

$$= \sum_{n_1=-\infty}^{\infty} x[n_1] e^{-j\omega(n_1+n_0)}$$

$$= + e^{-j\omega n_0} \sum_{n_1=-\infty}^{\infty} x[n_1] e^{-j\omega n_1}$$

$$= e^{-j\omega n_0} X(\omega)$$

Q

$$x[n] \leftrightarrow X(\omega)$$

$$x[-n] \leftrightarrow ?$$

$$F[x[-n]] = \sum_{n=-\infty}^{\infty} x[-n] e^{-j\omega n}$$

$$= \sum_{n_1=-\infty}^{\infty} x[n_1] e^{-j\omega(-n_1)} = \sum_{n_1=-\infty}^{\infty} x[n_1] e^{-j(-\omega)n_1}$$

$$= X(-\omega)$$

Q

$$x[n] \leftrightarrow X(\omega)$$

$$x^*[n] \leftrightarrow ?$$

$$F[x^*[n]] = \sum_{n=-\infty}^{\infty} x^*[n] e^{-j\omega n}$$

$$= \left[\sum x[n] e^{j\omega n} \right]^*$$

$$= \left[\sum x[n] e^{-j(-\omega)n} \right]^*$$

$$= X^*(-\omega)$$

Q

$$x[n] \leftrightarrow X(\omega)$$

$$n x[n] \leftrightarrow ?$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n}$$

$$\frac{dX(\omega)}{d\omega} = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} (-jn)$$

$$= -j \sum_{n=-\infty}^{\infty} n x[n] e^{-j\omega n}$$

$$-\frac{1}{j} \frac{dX(\omega)}{d\omega} = \sum_{n=-\infty}^{\infty} n x[n] e^{-j\omega n}$$

$$-\frac{1}{j} \times j \frac{dX(\omega)}{d\omega} = F[n x[n]]$$

$$F[n x[n]] = j \frac{dX(\omega)}{d\omega}$$

Q

$$x[n] = \delta[n] \quad \text{Find } X(\omega)$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} \delta[n] e^{-j\omega n}$$

$$X(\omega) = 1$$

Q

$$X(\omega) = 2\pi \delta(\omega - \omega_0) \quad ; \quad |\omega| \leq \pi \quad ; \quad |\omega_0| \leq \pi$$

$$\text{Find } x[n]$$

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\pi}^{\pi} 2\pi \delta(\omega - \omega_0) e^{j\omega n} d\omega$$

$$; (|\omega| \leq \pi)$$

$$= \frac{1}{2\pi} (2\pi (1) e^{j\omega_0 n})$$

$$\left(\begin{array}{l} @ \omega = \omega_0 \Rightarrow \delta(\omega - \omega_0) = 1 \\ \text{otherwise } \delta(\omega - \omega_0) = 0 \end{array} \right)$$

$$x[n] = e^{j\omega_0 n}$$

$$\& |\omega_0| \leq \pi$$

Q

$x[n] = 1$; Find $X(\omega)$ for $|\omega| \leq \pi$

$$e^{j\omega_0 n} \leftrightarrow 2\pi \delta(\omega - \omega_0)$$

Put $\omega_0 = 0$

$$e^0 \leftrightarrow 2\pi \delta(\omega)$$

$$1 \leftrightarrow 2\pi \delta(\omega)$$

$$\therefore X(\omega) = 2\pi \delta(\omega) \quad \text{for } |\omega| \leq \pi$$

$$X(\omega) = 2\pi \sum_{k=-\infty}^{\infty} \delta(\omega - 2\pi k)$$

for any value of ω
(Because $X(\omega)$ is periodic)
with period = 2π

Q

$x[n] = \cos \omega_0 n$; $|\omega_0| \leq \pi$. Find $X(\omega)$ for $|\omega| \leq \pi$

$$x[n] = \cos \omega_0 n = \frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2}$$

$$X(\omega) = \sum_{n=-\infty}^{\infty} \left(\frac{e^{j\omega_0 n} + e^{-j\omega_0 n}}{2} \right) e^{-j\omega n}$$

$$= \frac{1}{2} \left[\sum_{n=-\infty}^{\infty} e^{j\omega_0 n} e^{-j\omega n} + \sum_{n=-\infty}^{\infty} e^{-j\omega_0 n} e^{-j\omega n} \right]$$

$$= \frac{1}{2} \left[2\pi \delta(\omega - \omega_0) + 2\pi \delta(\omega + \omega_0) \right]$$

$$= \pi [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)] \quad \text{for } |\omega| \leq \pi$$

& $|\omega_0| \leq \pi$

For any value of ω

$$X(\omega) = \pi \sum_{k=-\infty}^{\infty} [\delta(\omega - \omega_0 - 2\pi k) + \delta(\omega + \omega_0 - 2\pi k)]$$

(Since $X(\omega)$ is periodic)
with period = 2π

Q $x[n] = u[n]$ Find $X(\omega)$ for $|\omega| \leq \pi$

$$u[n] = \frac{1}{2} [1 + \text{sgn}[n]]$$

$$\text{sgn}[n] - \text{sgn}[n-1] = 2\delta[n]$$

$$\text{Let } \text{sgn}[n] \longleftrightarrow X_1(\omega)$$

Taking Fourier Transform both sides

$$X_1(\omega) - X_1(\omega) e^{-j\omega} = 2(1)$$

$$X_1(\omega) = \frac{2}{1 - e^{-j\omega}}$$

$$F[u[n]] = F\left[\frac{1}{2} [1 + \text{sgn}[n]]\right]$$

$$= \frac{1}{2} [F[1] + F[\text{sgn}[n]]]$$

$$= \frac{1}{2} \left[2\pi\delta(\omega) + \frac{2}{1 - e^{-j\omega}} \right]$$

$$X(\omega) = \pi\delta(\omega) + \frac{1}{1 - e^{-j\omega}} \rightarrow \text{for } |\omega| \leq \pi$$

For any value of ω ,

$$X(\omega) = \frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi\delta(\omega - 2\pi k)$$

Q

$y[n] - \frac{1}{2} y[n-1] = x[n]$. Solve the difference equation
for input $x[n] = \left(\frac{1}{3}\right)^n u[n]$. Given that system was
initially relaxed i.e. $y[-1] = 0$

$$y[n] = y_{c.F.}[n] + y_{P.I.}[n]$$

$\Rightarrow$ For $y_{P.I.}[n] \rightarrow$ Forced response

$$\text{Let } y_{P.I.}[n] = A \left(\frac{1}{3}\right)^n \quad \text{for } n \geq 0$$

In steady state $y_{P.I.}[n]$ should satisfy the eq.
 $A \left(\frac{1}{3}\right)^n - \frac{1}{2} A \left(\frac{1}{3}\right)^{n-1} = \left(\frac{1}{3}\right)^n$

$$A - \frac{1}{2} A(3) = 1$$

$$A = -2$$

$$\Rightarrow \frac{2A}{2} - \frac{3A}{2} = 1$$

$$y_{P.I.} = -2 \left(\frac{1}{3}\right)^n$$

$\Rightarrow$ For $y_{c.F.} \rightarrow$ Force free response (Natural response)
 $y[n] - \frac{1}{2} y[n-1] = 0$ (Force free, $x[n]=0$)

$$y_{c.F.} = A_1 c^n$$

$$A_1 c^n - \frac{1}{2} A_1 c^{n-1} = 0$$

$$A_1 c^n \left[1 - \frac{1}{2} \frac{1}{c}\right] = 0 \quad \Rightarrow \quad \frac{1}{2c} = 1$$

$$c = \frac{1}{2}$$

$$y_{c.F.} = A_1 \left(\frac{1}{2}\right)^n$$

$$y[n] = y_{P.I.} + y_{c.F.} \quad ; \quad n \geq 0$$

$$y[n] = -2 \left(\frac{1}{3}\right)^n + A_1 \left(\frac{1}{2}\right)^n \quad ; \quad n \geq 0$$

For $n=0$: $y[0] - \frac{1}{2} y[-1] = \left(\frac{1}{2}\right)^0$

$$y[0] - \frac{1}{2}(0) = 1$$

($y[-1]=0$, Given)

$$y[0] = 1$$

$$y[n] = \left[-2 \left(\frac{1}{3}\right)^n + A_1 \left(\frac{1}{2}\right)^n \right] u[n]$$

@ $n=0$, $y[0] = 1$

$$1 = \left[-2 \left(\frac{1}{3}\right)^0 + A_1 \left(\frac{1}{2}\right)^0 \right] \cdot 1$$

$$A_1 = 3$$

$$\therefore y[n] = \left[-2 \left(\frac{1}{3}\right)^n + 3 \left(\frac{1}{2}\right)^n \right] u[n]$$

$$y[n] = \left[-2 \times \frac{2}{3} \left(\frac{1}{3}\right)^n + 3 \times \frac{2}{2} \left(\frac{1}{2}\right)^n \right] u[n]$$

$$y[n] = 6 \left[-\left(\frac{1}{3}\right)^{n+1} + \left(\frac{1}{2}\right)^{n+1} \right] u[n]$$

⇒ Discrete Fourier Transform (DFT), $X[k]$

→ Technique for Finite length Sequence

→ Discrete version of DTFT

→ Appropriate Fourier representation for digital Computer realization

→ One to One Correspondence between $x[n] \leftrightarrow X(k)$

N Point DFT

$$\left[\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi}{N}\right)kn} \\ x[n] &= \frac{1}{N} \sum_{k=0}^{N-1} X[k] e^{j\left(\frac{2\pi}{N}\right)kn} \end{aligned} \right. \quad ; k=0,1,2,3,\dots,N-1$$

→ Inverse DFT (IDFT)
 $n=0,1,2,3,\dots,N-1$

→ DTFT is periodic with periodicity of 2π
Frequency interval = $\frac{2\pi}{N}$

→ Choice of N is not fixed. If $x[n]$ has length $N_1 < N$, then $(N - N_1)$ Zeros must be added (Zero padding)

→ N is usually chosen to be a power of 2 (For Computational efficiency)

Q $x[n] = e^{-n}$, $0 \leq n \leq 4$. Compute N point DFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(\frac{2\pi}{N})kn}$$

$$\begin{aligned} N=5 \\ X[k] &= \sum_{n=0}^4 e^{-n} e^{-j(\frac{2\pi}{5})kn} = \sum_{n=0}^4 e^{-n[1 + j\frac{2\pi k}{5}]} \\ &= 1 + e^{-[1 + j\frac{2\pi k}{5}]} + \dots + e^{-4[1 + j\frac{2\pi k}{5}]} \\ &= \frac{1 - e^{-5(1 + j\frac{2\pi k}{5})}}{1 - e^{-(1 + j\frac{2\pi k}{5})}} = \frac{1 - e^{-5}}{1 - e^{-j(1 + j\frac{2\pi k}{5})}} \end{aligned}$$

⇒ For N greater than 5, Zero padding is required in $x[n]$

Q $x[n] = \begin{cases} 1 & 0 \leq n \leq N-1 \\ 0 & \text{otherwise} \end{cases}$. Compute N point DFT

$$\begin{aligned} \text{Case 1} \\ \text{For } k \neq 0 \\ X[k] &= \sum_{n=0}^{N-1} x[n] e^{-j(\frac{2\pi}{N})kn} = \sum_{n=0}^{N-1} (1) e^{-j(\frac{2\pi}{N})kn} \\ &= 1 + e^{-j(\frac{2\pi k}{N})} + \dots + e^{-j(\frac{2\pi k}{N})N-1} \\ &= \frac{1 - e^{-j(\frac{2\pi k}{N})N}}{1 - e^{-j(\frac{2\pi k}{N})}} = \frac{1 - e^{-j(2\pi k)}}{1 - e^{-j(\frac{2\pi k}{N})}} \\ &= 0 \end{aligned}$$

$$\text{Case 2} \\ \text{For } k=0 \\ X[k] = \sum_{n=0}^{N-1} 1 e^0 = N$$

Q $x[n] = \cos\left(\frac{\pi}{2}n\right)$, $n=0,1,2,3$. Find 4-point DFT

$$\begin{aligned} X[k] &= \sum_{n=0}^3 \cos\left(\frac{\pi}{2}n\right) e^{-j\left(\frac{2\pi k}{4}\right)n} \\ &= 1 + 0 + (-1)e^{-j\left(\frac{2\pi k}{4}\right)2} + 0 \\ &= 1 - e^{-j\pi k}, \quad k=0,1,2,3 \end{aligned}$$

Q $x[n] = a^n$, $0 \leq n \leq N-1$. Compute DFT

$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi k}{N}\right)n}, \quad k=0,1,\dots,N-1 \\ &= \sum_{n=0}^{N-1} a^n e^{-j\left(\frac{2\pi k}{N}\right)n} \\ &= 1 + a e^{-j\left(\frac{2\pi k}{N}\right)} + \dots + a^{N-1} e^{-j\left(\frac{2\pi k}{N}\right)(N-1)} \\ &= \frac{1 - \left(a e^{-j\left(\frac{2\pi k}{N}\right)}\right)^N}{1 - a e^{-j\left(\frac{2\pi k}{N}\right)}} = \frac{1 - a^N e^{-j(2\pi k)}}{1 - a e^{-j\left(\frac{2\pi k}{N}\right)}} \end{aligned}$$

$$X[k] = \frac{1 - a^N}{1 - a e^{-j\left(\frac{2\pi k}{N}\right)}}, \quad k=0,1,2,\dots,N-1$$

$\Rightarrow$ Fast Fourier Transform (FFT)

- Class of algorithms for efficient computing DFT.
- Decompose successively the N-point DFT computation into smaller size DFT computation.
- Algorithm results in significant saving in computational complexity.
- 2 Basic class of FFT algorithm
 - $\rightarrow$ Decimation in Time FFT
 - $\rightarrow$ Decimation in Frequency FFT

⇒ Decimation in time FFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j\left(\frac{2\pi}{N}\right)kn} \rightarrow N \text{ point DFT}$$

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} \quad \left(\text{where } W_N = e^{-j\left(\frac{2\pi}{N}\right)} \right)$$

$$\begin{aligned} X[k] &= \sum_{n=\text{even}} x[n] W_N^{kn} + \sum_{n=\text{odd}} x[n] W_N^{kn} \\ &= \sum_{n=0}^{\frac{N}{2}-1} x[2n] W_N^{k(2n)} + \sum_{n=0}^{\frac{N}{2}-1} x[2n+1] W_N^{k(2n+1)} \\ &= \sum_{n=0}^{\frac{N}{2}-1} x[2n] W_{\frac{N}{2}}^{kn} + \sum_{n=0}^{\frac{N}{2}-1} x[2n+1] W_{\frac{N}{2}}^{kn} W_N^k \end{aligned}$$

$$\begin{aligned} W_N^{k+\frac{N}{2}} &= W_N^k W_N^{\frac{N}{2}} \\ &= W_N^k e^{-j\left(\frac{2\pi}{N}\right)\frac{N}{2}} \\ &= W_N^k e^{-j\pi} \\ &= -W_N^k \end{aligned}$$

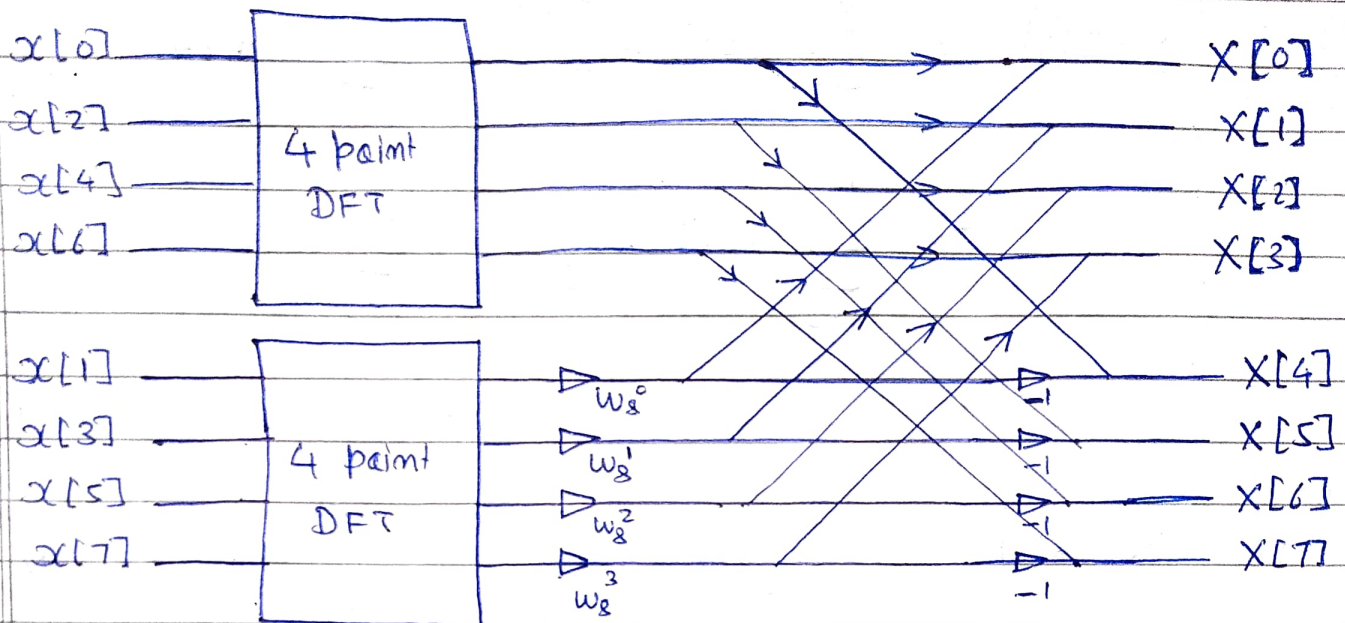
$$\begin{aligned} W_{\frac{N}{2}}^{\frac{N}{2}n} &= e^{-j\left(\frac{2\pi}{\frac{N}{2}}\right)\frac{N}{2}n} \\ &= e^{-j(2\pi)n} \\ &= 1 \end{aligned}$$

For $k=0, 1, 2, \dots, \frac{N}{2}-1$

$$\begin{aligned} X[k] &= \sum_{n=0}^{\frac{N}{2}-1} x[2n] W_{\frac{N}{2}}^{kn} + \left(\sum_{n=0}^{\frac{N}{2}-1} x[2n+1] W_{\frac{N}{2}}^{kn} \right) W_N^k \\ X\left[k+\frac{N}{2}\right] &= \sum_{n=0}^{\frac{N}{2}-1} x[2n] W_{\frac{N}{2}}^{(k+\frac{N}{2})n} + \sum_{n=0}^{\frac{N}{2}-1} x[2n+1] W_{\frac{N}{2}}^{(k+\frac{N}{2})n} W_N^{k+\frac{N}{2}} \\ &= \sum_{n=0}^{\frac{N}{2}-1} x[2n] W_{\frac{N}{2}}^{kn} (1) + \sum_{n=0}^{\frac{N}{2}-1} x[2n+1] W_{\frac{N}{2}}^{kn} (1) (-W_N^k) \end{aligned}$$

Decomposition of N point DFT into $\frac{N}{2}$ point DFT

8 point - Decimation in time FFT



Decimation in Frequency FFT

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)kn} = \sum_{n=0}^{N-1} x[n] W_N^{kn}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{kn} + \sum_{n=\frac{N}{2}}^{N-1} x[n] W_N^{kn}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{kn} + \sum_{m=0}^{\frac{N}{2}-1} x[m+\frac{N}{2}] W_N^{k(m+\frac{N}{2})}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{kn} + \sum_{m=0}^{\frac{N}{2}-1} x[m+\frac{N}{2}] W_N^{km} W_N^{k\frac{N}{2}} \quad \left(n \rightarrow m + \frac{N}{2} \right)$$

$$\left(W_N^{k\frac{N}{2}} = e^{-j(2\pi/N)k\frac{N}{2}} = e^{-j\pi k} = (-1)^k \right)$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{kn} + (-1)^k \sum_{n=0}^{\frac{N}{2}-1} x[n+\frac{N}{2}] W_N^{kn} \quad (m \rightarrow n)$$

$$X[2K_1] = \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{2K_1 n} + \sum_{n=0}^{\frac{N}{2}-1} x[n+\frac{N}{2}] W_N^{2K_1 n} (-1)^{2K_1}$$

$$X[2K_1] = \sum_{n=0}^{\frac{N}{2}-1} x[n] W_{\frac{N}{2}}^{K_1 n} + \sum_{n=0}^{\frac{N}{2}-1} x[n+\frac{N}{2}] W_{\frac{N}{2}}^{K_1 n}$$

$$K_1 = 0, 1, 2, \dots, \frac{N}{2}-1$$

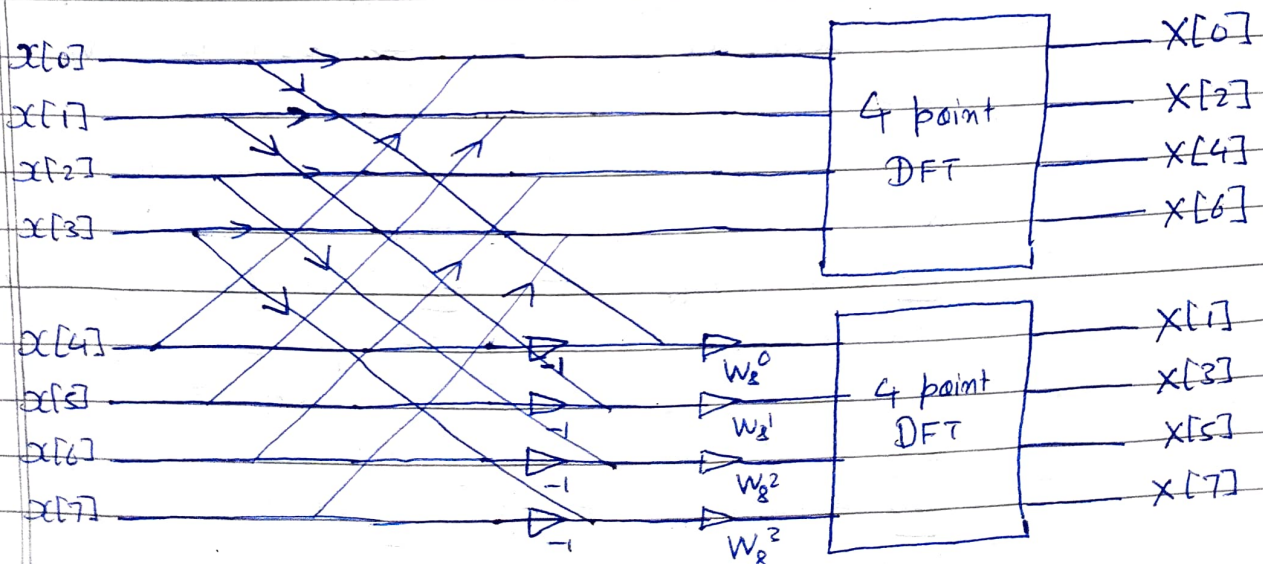
$$X[2K_1+1] = \sum_{n=0}^{\frac{N}{2}-1} x[n] W_N^{(2K_1+1)n} + \sum_{n=0}^{\frac{N}{2}-1} x[n+\frac{N}{2}] W_N^{(2K_1+1)n} (-1)^{2K_1+1}$$

$$X[2K_1+1] = \sum_{n=0}^{\frac{N}{2}-1} x[n] W_{\frac{N}{2}}^{K_1 n} W_N^n - \sum_{n=0}^{\frac{N}{2}-1} x[n+\frac{N}{2}] W_{\frac{N}{2}}^{K_1 n} W_N^n$$

$$K_1 = 0, 1, 2, 3, \dots, \frac{N}{2}-1$$

⇒ Decomposition of N point DFT into $\frac{N}{2}$ point DFT

8 point Decimation in frequency FFT algorithm



Q $x[n] = \{1, 1, -1, -1, -1, 1, 1, -1\}$. Find 8 point DFT using decimation in time FFT algorithm

$$\left. \begin{array}{l} \text{Even Seq. : } x[2n] = \{1, -1, -1, 1\} \\ \text{Odd Seq. : } x[2n+1] = \{1, -1, 1, -1\} \end{array} \right\} n = 0, 1, 2, 3$$

$\Rightarrow$ 4 point DFT₃ of $x[2n]$:

$$\begin{aligned} X_E[k] &= \sum_{n=0}^3 x[2n] e^{-j(\frac{2\pi}{4})kn} \\ &= (1) e^0 + (-1) e^{-j(\frac{2\pi}{4})k(1)} + (-1) e^{-j(\frac{2\pi}{4})k(2)} + (1) e^{-j(\frac{2\pi}{4})k(3)} \\ &= 1 + (-1)(-j)^k + (-1)(-1)^k + (j)^k \end{aligned}$$

$$X_E[0] = 1 - 1 - 1 + 1 = 0$$

$$X_E[1] = 1 + j + 1 + j = 2 + 2j$$

$$X_E[2] = 1 + (-1)(-1) + (-1)(1) + (-1) = 0$$

$$X_E[3] = 1 + (-1)(j) + (-1)(-1) - j = 2 - 2j$$

$\Rightarrow$ 4 point DFT₃ of $x[2n+1]$:

$$\begin{aligned} X_O[k] &= \sum_{n=0}^3 x[2n+1] e^{-j(\frac{2\pi}{4})kn} \\ &= 1 + (-1)(-j)^k + (1)(-1)^k + (-1)j^k \end{aligned}$$

$$X_O[0] = 1 - 1 + 1 - 1 = 0$$

$$X_O[1] = 1 + j + (1)(-1) + (-1)j = 0$$

$$X_O[2] = 1 + (-1)(-1) + (1)(1) + (-1)(-1) = 4$$

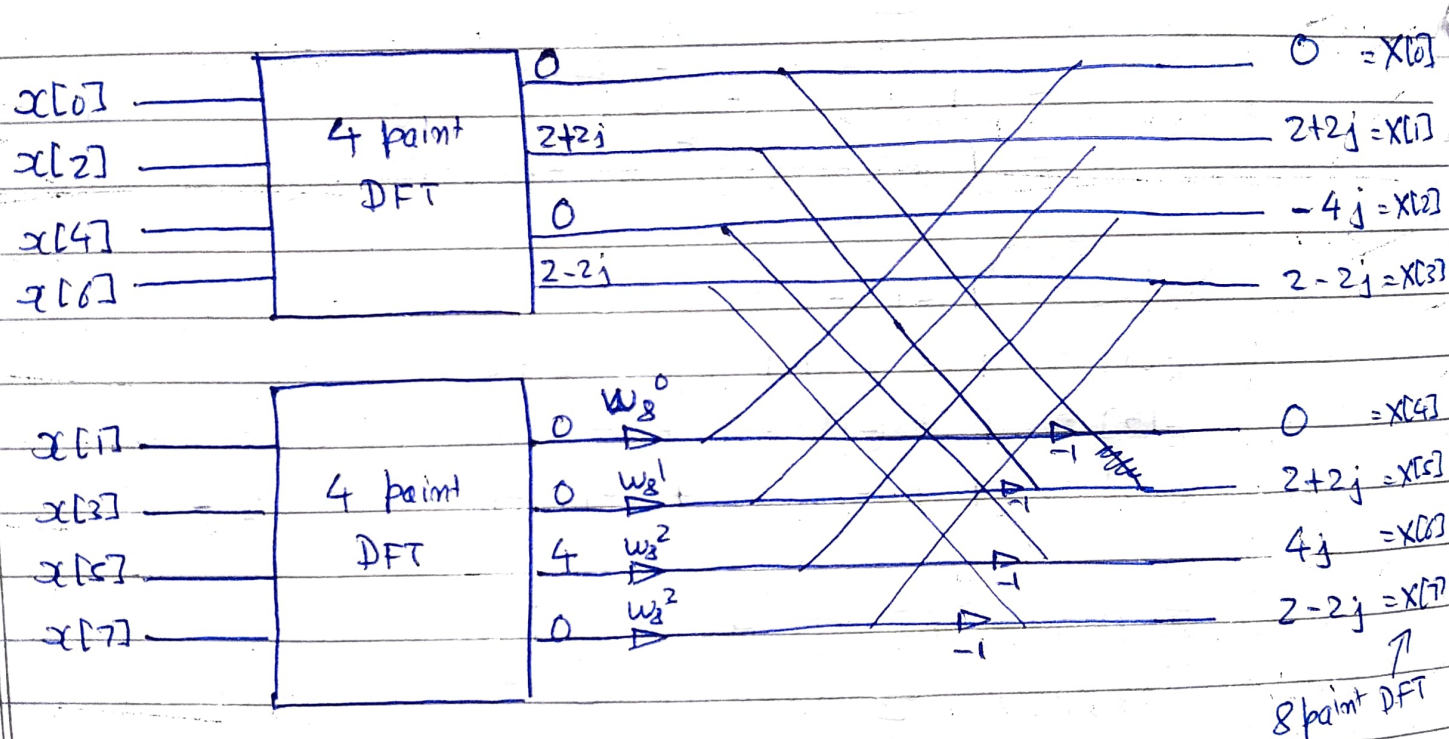
$$X_O[3] = 1 - j + (1)(-1) + j = 0$$

$$W_8^0 = e^{-j\left(\frac{2\pi}{8}\right)0} = 1$$

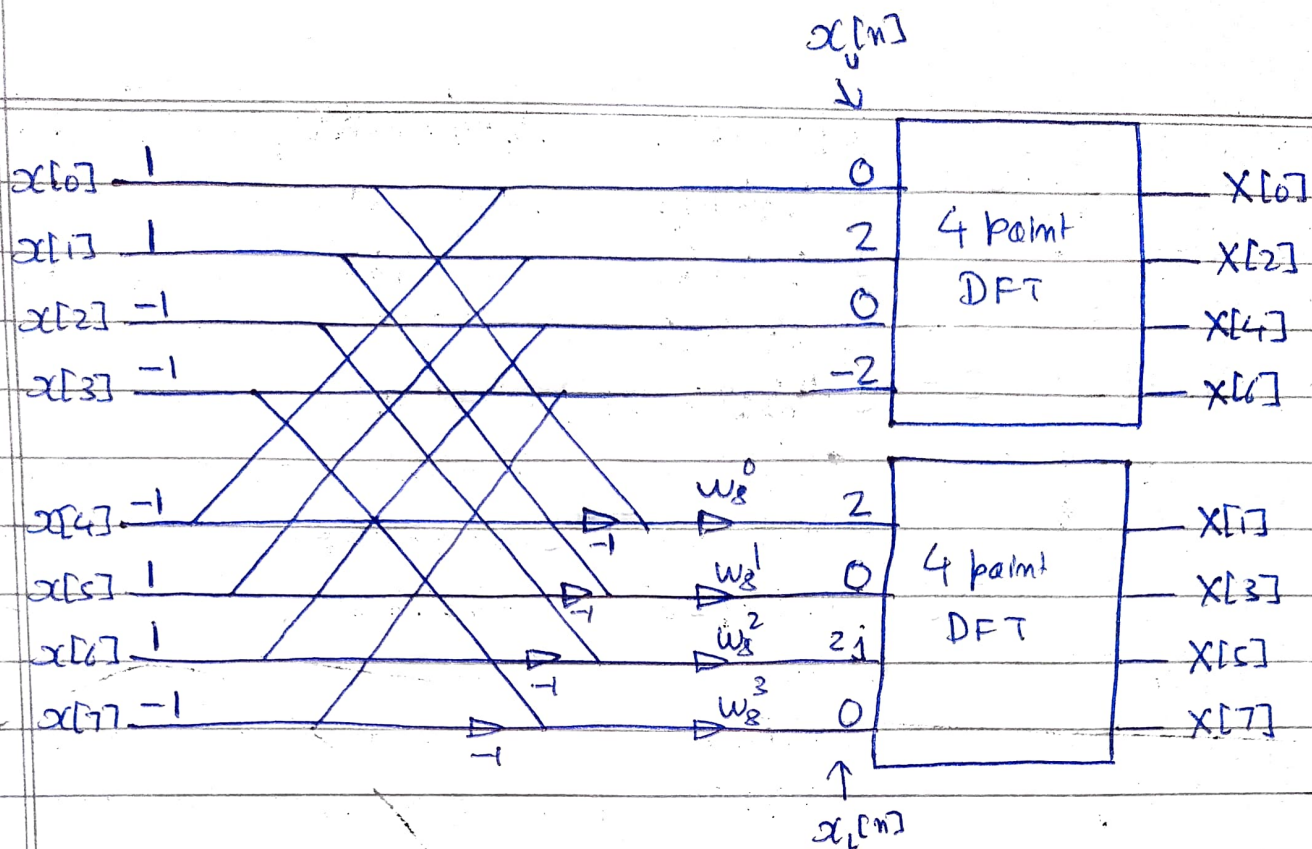
$$W_8^1 = e^{-j\left(\frac{2\pi}{8}\right)1} = e^{-j\frac{\pi}{4}} = \frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$$

$$W_8^2 = e^{-j\left(\frac{2\pi}{8}\right)2} = e^{-j\frac{\pi}{2}} = -j$$

$$W_8^3 = e^{-j\left(\frac{2\pi}{8}\right)3} = -\frac{1}{\sqrt{2}} - j\frac{1}{\sqrt{2}}$$



Q $x[n] = \{1, 1, -1, -1, -1, 1, 1, -1\}$. Compute 8 point DFT using decimation in frequency FFT algorithm.



$$x_u[n] = \{0, 2, 0, -2\} \rightarrow \text{upper Seq.}$$

$$x_l[n] = \{2, 0, 2j, 0\} \rightarrow \text{Lower Seq.}$$

⇒ 4 point DFT of upper sequence, $x_u[n]$

$$X[2k] = \sum_{n=0}^3 x_u[n] e^{-j\left(\frac{2\pi}{4}\right)kn} \quad ; k=0,1,2,3$$

$$= 0(+1) + 2(-j)^k + 0 + (-2)(j)^k$$

$$X[0] = 0$$

$$X[2] = 2(-j)^1 + (-2)(j)^1 = -4j$$

$$X[4] = 2(-j)^2 + (-2)(j)^2 = 0$$

$$X[6] = 2(-j)^3 + (-2)(j)^3 = 4j$$

⇒ 4 point DFT of Lower Sequence, $x_l[n]$

$$X[2k+1] = \sum_{n=0}^3 x_l[n] e^{-j\left(\frac{2\pi}{4}\right)kn}$$

$$= 2(1) + 0 + 2j(-1)^k + 0$$

$$X[1] = 2 + 2j$$

$$X[3] = 2 - 2j$$

$$X[5] = 2 + 2j$$

$$X[7] = 2 - 2j$$

$$X[k] = \left\{ \begin{array}{l} 0, 2+2j, -4j, 2-2j, 0, 2+2j, \\ 4j, 2-2j \end{array} \right\}$$

8 point DFT
of $x[n]$